

Grid-Forming Inverter Applications for Improved Reliability and Power Quality in AI Data Centers

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Abstract—Based on the rapid growth in internet-based technology and the prominence of artificial intelligence (AI), data centers are becoming a critical energy infrastructure in today's power system. Their continuous, short-term, and intermittent load profiles lead to power quality issues that can degrade the network's reliability. Grid-forming (GFM) inverters have emerged as a promising solution to various challenges posed by the growing integration of inverter-based resources with grid-following characteristics. In this paper, we examine the impacts of voltage sag and flicker on power systems with AI data centers as significant loads and propose the adoption of GFM inverters to address these power quality issues.

Index Terms—Data center, flicker, grid-forming inverter, power quality, voltage ride-through

I. INTRODUCTION

Emerging power systems are evolving quickly, driven by environmental goals and advances in renewable technologies, particularly the increased integration of renewable energy sources and power electronics. The integration of renewable resources, such as solar and wind, introduces variability and uncertainty, undermining the predictability of fossil-fuel-based generation [1]. High penetration of power electronics, including devices like converters for wind and solar, HVDC transmission, and electric vehicle chargers, offers greater control over power flow, enhancing efficiency but also creating unique challenges in system stability and reliability. These converter-dominated grids introduce low inertia, complex electromagnetic dynamics, and multi-timescale control interactions that complicate conventional stability models, which were designed for traditional synchronous generators. Consequently, addressing these challenges necessitates innovative control strategies, robust fault management, and adaptable stability frameworks to maintain reliable operation amidst an increasingly decentralized and renewable-heavy power system [2].

The rapid expansion of data centers to meet increasing digital demands has created several challenges related to power system integration, especially concerning power quality and energy efficiency. Data centers require stable, high-quality power to support their sensitive electronic equipment, yet their high energy consumption introduces issues like voltage sags, harmonic distortion, and frequent power transients [3]. Even minor fluctuations in voltage or frequency can disrupt operations, causing data loss, equipment malfunction, and significant financial impact. To counteract these disturbances, power conditioning solutions, such as Uninterruptible Power

Supplies (UPS) and specialized filtering techniques, are essential, ensuring consistent power delivery and minimizing operational risks [4].

Conventional power quality improvement methods frequently rely on devices like Static Var Compensators (SVCs), Static Synchronous Compensators (STATCOMs), and Dynamic Voltage Restorers (DVRs) to address issues such as voltage sags, swells, and harmonic distortion. SVCs, which use thyristor-controlled reactors and capacitors, are widely used for reactive power compensation and voltage stabilization, but they come with a high cost due to their need for constant maintenance, high-voltage switching technology, and additional harmonic filters. STATCOMs, which are voltage-source converter-based systems, offer improved dynamic performance and faster response times than SVCs, making them highly effective for voltage support; however, their advanced technology and power electronics components lead to high installation and operational costs [5]. DVRs, on the other hand, provide rapid voltage restoration during sags and swells by injecting compensating voltage into the system, which protects sensitive loads. However, like SVCs and STATCOMs, DVRs are also costly, requiring sophisticated control systems, large transformers, and regular upkeep to ensure reliable performance. While these solutions effectively enhance power quality, the high initial and maintenance expenses associated with each device often pose financial challenges, especially for applications demanding large-scale or continuous compensation.

The concept of grid-forming inverters (GFM inverters) has emerged as a powerful tool for stabilizing modern power systems, especially under high renewable penetration. Unlike traditional grid-following (GFL) inverters, which synchronize to the grid's voltage and frequency and operate as current sources, GFM inverters function as voltage sources with the capability to set and maintain their own voltage and frequency parameters. This voltage-source characteristic allows GFM inverters to operate independently of grid conditions and provide critical support in scenarios such as islanded operation or weak grid conditions [6].

One of the main advantages of GFM inverters over GFL inverters is their ability to offer inertia-like behavior, which supports grid frequency stability and enhances voltage control during disturbances, providing a level of robustness typically associated with synchronous generators. GFM inverters, through advanced control strategies like virtual synchronous

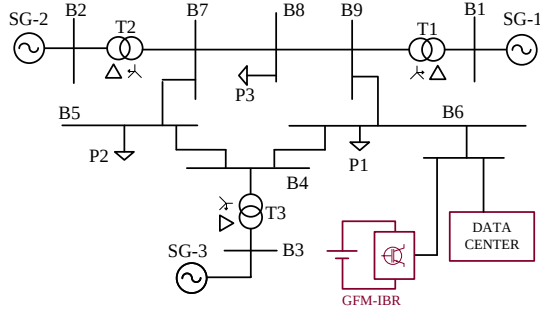


Fig. 1. IEEE 9-bus system with a grid-connected data center and a GFM inverter.

machine (VSM) and droop control, can help mitigate frequency and voltage instabilities, especially in grids with low inertia where conventional inverters fall short [7]. Furthermore, by stabilizing the voltage, GFM inverters mitigate risks associated with power quality issues and protecting sensitive data center equipment from outages or malfunctions [8].

This paper models a data center integrated with a GFM in a 9-bus IEEE system to demonstrate its effectiveness in maintaining grid stability and power quality during faults or islanding. GFM inverters' unique voltage-forming capability allows them to remain operational during low-voltage conditions, providing essential support for voltage stability and reactive power, crucial for sensitive systems like data centers. Additionally, the study investigates GFM inverters' role in mitigating voltage flicker and improving frequency regulation during islanding, showcasing their advanced control strategies and inertia-like behavior. These contributions highlight the GFM's potential to address key challenges in modern power systems with high renewable integration.

The rest of this paper is organized as follows. Section II describes the model of elements of the system architecture and flicker measurement methodology. Section III presents the simulation setup and discusses the results. Section IV concludes the findings, highlighting the role of GFM-controlled battery energy storage systems (BESSs) in enhancing power quality and system reliability in AI-driven data centers.

II. SYSTEM ARCHITECTURE AND FLICKER MEASUREMENT METHODOLOGY

Appropriate models need to be developed to investigate the impacts of voltage disturbances on data centers. In this section, the system architecture used for simulation is presented. The schematic is shown in Fig. 1. It depicts the IEEE 9-bus system [9], representing a typical interconnected power network, a data center, and a GFM-based BESS. The following subsections provide a detailed description of the data center and GFM inverter modeling.

A. Data Center Modeling

A typical grid-connected data center exhibits GFL characteristics, consisting of multiple racks ranging from tens to hundreds, and each rack has multiple vaults (hundreds).

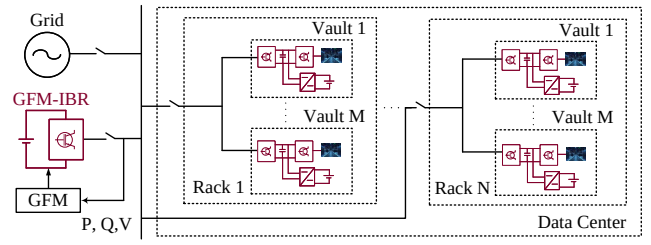


Fig. 2. Grid-connected data center with grid-forming inverter

Each vault is a complex system comprising several power converters, as shown in Fig. 2 [10]. The rectifier converts the input AC voltage to DC; the DC-DC converter is an interface between the DC bus and the battery. The inverter transforms the DC power into AC to serve the IT and mechanical loads. The function of the battery interface is to maintain the DC link voltage within specified limits to sustain power transfer to the load for a certain period.

When a voltage disturbance event occurs in the system and the point of interconnection (POI) voltage is out of specified limits, the loads are disconnected from the system. This sudden disconnection of large AI data center loads further exacerbates the system's reliability.

Additionally, modern data centers have base loads (usually 35% of the peak load) and AI loads that behave differently and can impact the system's reliability. The load profile of a typical data center, shown in Fig. 3, can be written as:

$$L(t) = \begin{cases} L_b & \text{for } 0 \leq t < t_a \\ L_b + \frac{L_{pk} - L_b}{t_b - t_a} \cdot (t - t_a) & \text{for } t_a \leq t < t_b \\ L_{pk} & \text{for } t_b \leq t < t_c \\ L_{pk} + \frac{L_b - L_{pk}}{T - t_c} \cdot (T - t) & \text{for } t_c \leq t < T \end{cases} \quad (1)$$

where L_b is the base load, L_{pk} is the peak load, t_a is duration of the rising edge, t_b is the time to reach peak load, t_c is time for the falling edge, and T is the period of the signal. This periodic behavior of the data center at varying frequencies can lead to another power quality issue known as flicker in the system, thereby compromising the system's reliability. It becomes imperative that modern data centers comply with IEEE 1453-2022.

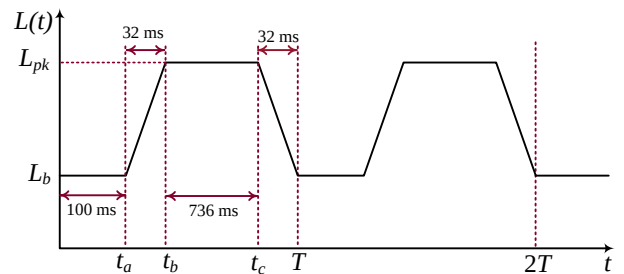


Fig. 3. Periodic load profile of a typical AI-based data center

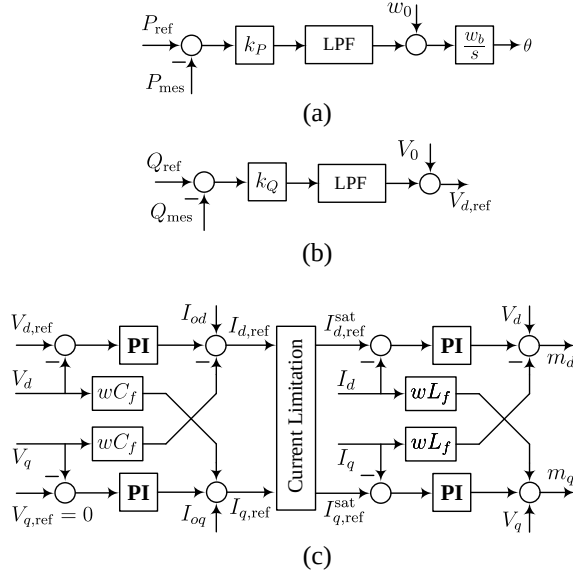


Fig. 4. GFM control (a) Active power droop control (b) Reactive power droop controller (c) Inner voltage and current controllers [12].

B. Grid-forming Inverter Modeling

In contrast to GFL inverters, a GFM inverter control topology actively manages both voltage and frequency at the POI, causing the inverter to emulate a voltage source. This design enables the GFM inverter to seamlessly operate in both grid-connected and islanded modes, providing the necessary flexibility to handle fluctuations in load independently when in islanded mode [11].

In this paper, a droop-based control approach is implemented for the GFM inverter to balance load sharing between the GFM and a synchronous generator. The droop-based control methodology adjusts the inverter's frequency and voltage in response to active and reactive power, respectively. This strategy allows the GFM inverter to adapt its power output based on system demands, contributing to stable operation alongside the synchronous generator in both grid-connected and islanded scenarios. The active power droop control, as shown in Fig. 4(a), adjusts the frequency (ω) according to the active power (P) output of the inverter, defined by the relationship:

$$\omega = \omega_0 - k_P \cdot (P_{mes} - P_{ref}) \quad (2)$$

where ω_0 is the nominal frequency, P_{ref} is the reference active power, and k_P is the droop coefficient that determines the sensitivity of frequency to changes in active power. To enhance stability and mitigate high-frequency oscillations, a low-pass filter (LPF) is incorporated, resulting in the following transfer function for active power control:

$$G_P(s) = k_P \frac{\omega_c}{s + \omega_c} \quad (3)$$

where ω_c represents the cut-off frequency of the LPF.

Similarly, reactive power droop control operates based on the linear dependency between voltage magnitude (V) at the POI and the reactive power (Q) managed by the inverter, illustrated in Fig. 4(b). The reactive power droop equation is:

$$V = V_0 - k_Q \cdot (Q_{mes} - Q_{ref}) \quad (4)$$

where V_0 is the nominal voltage, Q_{ref} is the reference reactive power, and k_Q is the droop coefficient for reactive power, adjusted to balance reactive power among parallel inverters in the system. The LPF applied in the reactive power controller follows the transfer function:

$$G_Q(s) = k_Q \frac{\omega_c}{s + \omega_c} \quad (5)$$

In addition to the droop control loops, inner voltage and current controllers are essential for stable GFM inverter operation, as shown in Fig. 4(c). These inner control loops can be described using the proportional-integral (PI) control structure [13]. The PI controllers for the voltage and current loops are defined by:

$$G_v(s) = K_{p-v} + \frac{1}{T_{i-v}s} \quad (6)$$

$$G_i(s) = K_{p-i} + \frac{1}{T_{i-i}s} \quad (7)$$

where K_{p-v} and T_{i-v} represent the proportional gain and integral time constant of the PI controller for the voltage control loop, respectively, and K_{p-i} and T_{i-i} represent the proportional gain and integral time constant for the current control loop [12]. These parameters are critical for ensuring stable control and effective response under dynamic conditions in the GFM inverter.

The current saturation scheme is applied to ensure that the current reference values do not exceed the maximum allowable current, I_{max} , thus preventing potential overcurrent issues [14], [15]. This limitation is achieved by defining the saturated current reference, i_{dq-ref}^{sat} , as:

$$i_{dq-ref}^{sat} = \begin{cases} i_{dq-ref} & \text{if } \sqrt{i_{d-ref}^2 + i_{q-ref}^2} \leq I_{max} \\ \rho i_{dq-ref} & \text{if } \sqrt{i_{d-ref}^2 + i_{q-ref}^2} > I_{max} \end{cases} \quad (8)$$

where i_{dq-ref} is the current reference in the dq -frame, and ρ is a scaling factor applied to limit the current to I_{max} when the combined dq -axis current reference exceeds this threshold. This ensures that the current remains within safe operating limits.

With this droop-based control architecture, the GFM inverter not only provides voltage and frequency support at the interconnection point but also dynamically adapts to changes in active and reactive power, crucial for effective load sharing and grid stability [16].

C. Flicker Measurement Methodology

In compliance with IEEE 1453-2022, flicker severity was measured using a flicker meter that models human sensitivity

to light fluctuations. Short-term flicker severity, P_{st} , calculated over 10 minutes, is given by:

$$P_{st} = \sqrt{\left(0.0314P_{0.1} + 0.0525P_1 + 0.0657P_3 + 0.28P_{10} + 0.08P_{50}\right)} \quad (9)$$

where $P_{0.1}$, P_1 , P_3 , P_{10} , and P_{50} represent flicker levels exceeded for 0.1%, 1%, 3%, 10%, and 50% of the time, respectively. Per the IEC61000-3-7:2008 standard, flicker limits are $P_{st} \leq 0.9$, $P_{lt} \leq 0.7$ for medium voltage and $P_{st} \leq 0.8$, $P_{lt} \leq 0.6$ for high voltage, ensuring compliance with electromagnetic compatibility standards.

III. SIMULATION RESULTS AND DISCUSSION

This section outlines the results of the PSCAD simulation of the study system presented in Fig 1, where the data center has a peak load of 500 MW, and the droop-based BESS-GFM is rated at 200 MW. The following elaborates on two scenarios considered in this study.

A. Voltage Sag Mitigation with Grid-Forming Inverter

In this scenario, the impact of low-voltage events on the system's reliability is investigated with and without the BESS-GFM system. With the increasing penetration of large loads such as data centers, any disconnection in the load due to voltage events will significantly impact the load-generation balance in the grid. A single line-to-ground fault is applied at Bus B8 at $t = 4$ s, and cleared after 6 cycles. The comparative voltage and current profiles of this scenario are shown in Fig. 5. The "Fault inception" is expected to cause a voltage sag (below 0.9 pu) in the system as shown in Fig. 5(a)(i)–purple plot. To circumvent further complications in the AI data center connected to the grid and ensure compliance with operational standards, when the voltage drops below 0.9 pu, the data center trips as shown in Fig. 5(b). The disconnection of the data center causes a voltage rise at the POI, keeping the data center within the operating limits, as shown in Fig. 5(a)(ii)–green plot.

When the BESS-GFM is connected to the system, it is observed that the voltage profile at the POI is above the 0.9-pu threshold and within nominal operating limits, as shown in Fig. 5(a)(iii)–blue plot. This ride-through capability ensures that the AI data center remains connected to the system as depicted Fig. 5(c). Additionally, Fig. 6 shows the frequency of the system with and without BESS-GFM. It is seen that the frequency of the system is within steady-state operating conditions when the BESS-GFM is connected, maintaining the load-generation balance in the system.

B. Flicker Mitigation with Grid-Forming Inverter

AI data centers present unique challenges to power quality, especially in terms of flicker, due to their nonlinear load characteristics and rapidly fluctuating load profiles. As illustrated by Fig. 7(a), the high variability of the data center's load

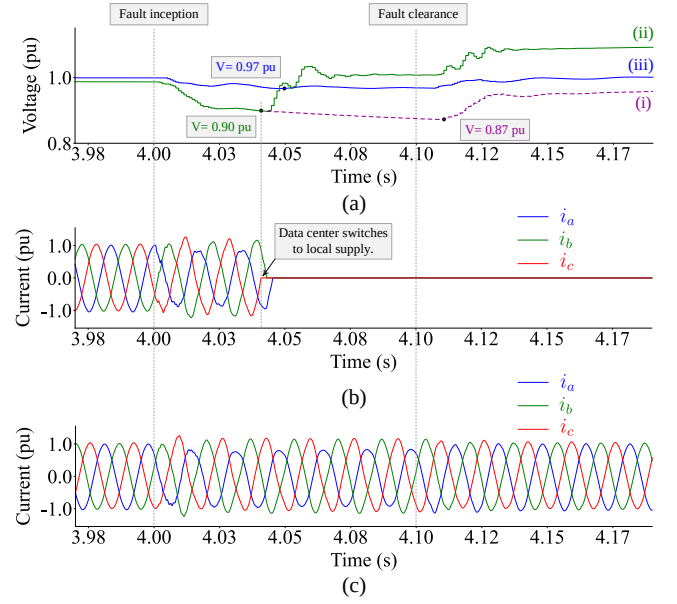


Fig. 5. Measurements at the POI of the AI data center, (a) voltage magnitude, (b) current without BESS-GFM, (c) current with BESS-GFM.

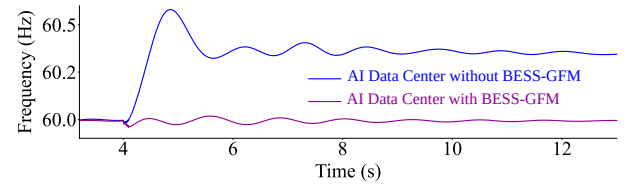


Fig. 6. System frequency with and without BESS-GFM

results in notable voltage flicker. These flicker values pose a risk to system stability and could affect the reliability of sensitive equipment connected to the grid.

To address this issue, a solution was implemented by integrating a BESS-GFM at the same bus (Bus B6) in the IEEE 9-bus system. The BESS-GFM, operating as a voltage source with robust control capabilities, effectively compensates for the voltage fluctuations induced by the data center's variable load. This configuration stabilizes the power quality at Bus B6 by dynamically adjusting voltage and frequency in response to load changes.

As illustrated in Fig. 7(b), after connecting the BESS-GFM, there is a significant reduction in flicker level compared to the initial scenario. As shown by Fig. 8, the probability and cumulative density functions of the instantaneous flicker levels shift towards lower values, indicating that the magnitude and frequency of flicker have been mitigated.

Quantitatively, the short-term flicker severity (P_{st}) for both scenarios was evaluated, with and without the BESS-GFM, as summarized in Table I. The comparison shows a substantial difference in flicker severity, with a 98.93% reduction achieved by integrating the BESS-GFM.

Fig. 7 illustrates the improvement in flicker mitigation at the POI with the addition of the BESS-GFM. Fig. 7 (a),

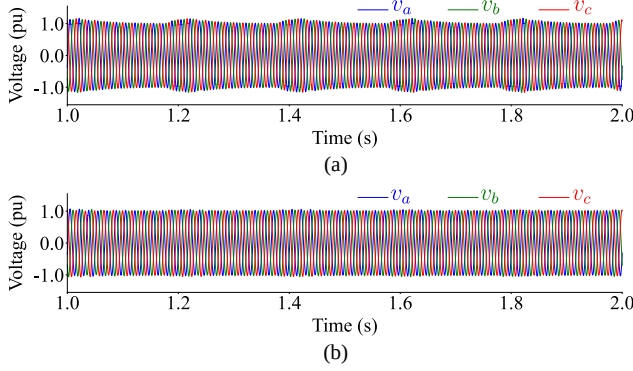


Fig. 7. POI voltage (a) without BESS-GFM, (b) with BESS-GFM.

TABLE I
 P_{st} LEVELS WITH AND WITHOUT BESS-GFM

SCR	P_{st} level without BESS-GFM	P_{st} level with BESS-GFM
High	0.2061	0.0212
Low	0.947	0.278

representing the scenario without the BESS-GFM, the voltage waveform shows noticeable oscillation due to flicker caused by the rapidly fluctuating load of the AI-driven data center. These fluctuations contribute to a high flicker severity level, impacting power quality. In contrast, Fig. 7 (b) shows the scenario with the BESS-GFM connected, where the voltage waveform is significantly smoother. This stability reflects the BESS-GFM's capability to mitigate flicker by compensating for load-induced voltage oscillations, thus improving power quality and reducing the flicker severity at the POI. It is worth mentioning that factors like system strength, magnitude and frequency of load variations, and response time of inverter controls can impact the BESS-GFM requirements for flicker mitigation.

IV. CONCLUSION

This paper demonstrated that integrating BESS-GFM with AI-driven data centers effectively enhances power quality and system reliability. Simulations on a modified IEEE 9-bus system showed that BESS-GFM mitigates voltage sags and flicker caused by data centers' fluctuating loads, ensuring stable voltage and frequency even during low-voltage events. By maintaining connection and reactive support under disturbances, BESS-GFM prevents load disconnections and supports IEEE power quality standards, making them a valuable solution for stability in systems with high inverter-based and renewable resources.

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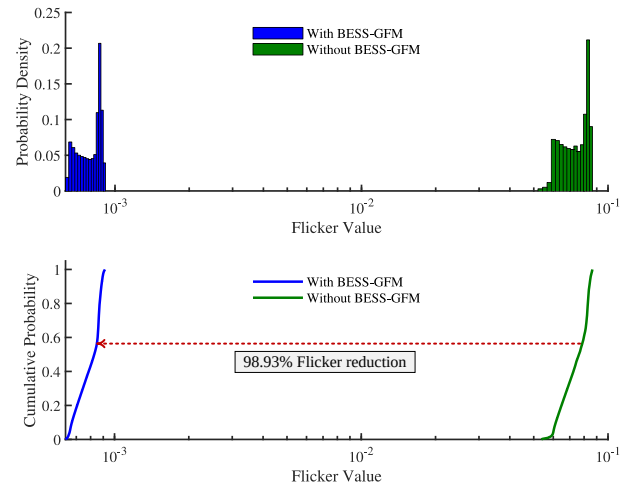


Fig. 8. Probability density and cumulative probability of flicker values with and without BESS-GFM, plotted on a logarithmic x-axis for better visualization.

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